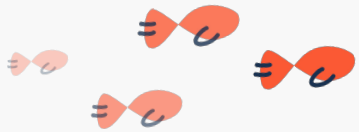


School and Pool for Digital Talent



A picture is worth a thousand words...

Telling our stories through graphs and charts!

Why do we need visualisation?

Because it helps us tell our stories.

As a data scientist, you need to help others understand the insight that your data provides.

To get to that, let's start with a series of mini exercises:



A picture is worth a thousand words...

Telling our stories through graphs and charts!
Exercise One:

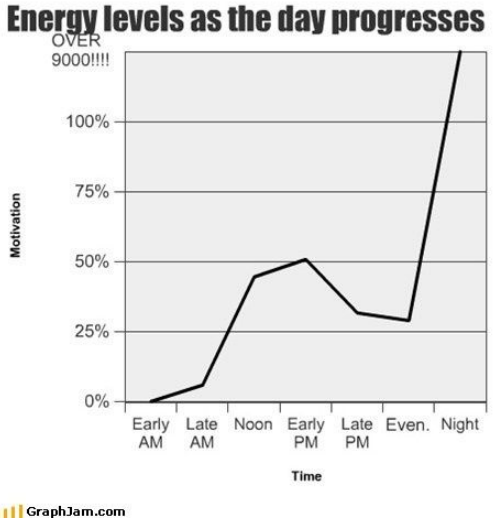
Think of something in your life create a graph or a chart to explain it to others.

Later on, you'll need to explain:

- what you visualised
- why in this way

And

- key messages/findings

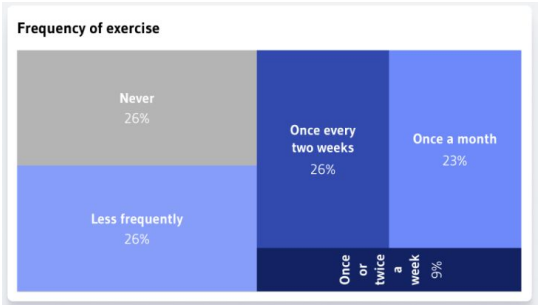


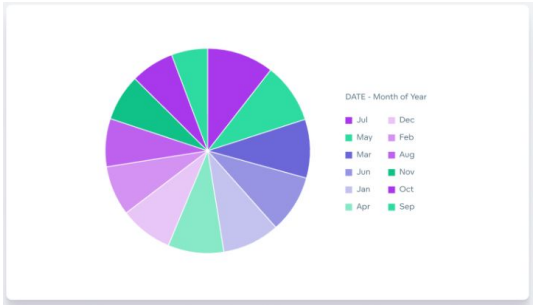
A picture is worth a thousand words...

Telling our stories through graphs and charts!
Exercise Two:

None of these is good visualization. What's wrong with them?





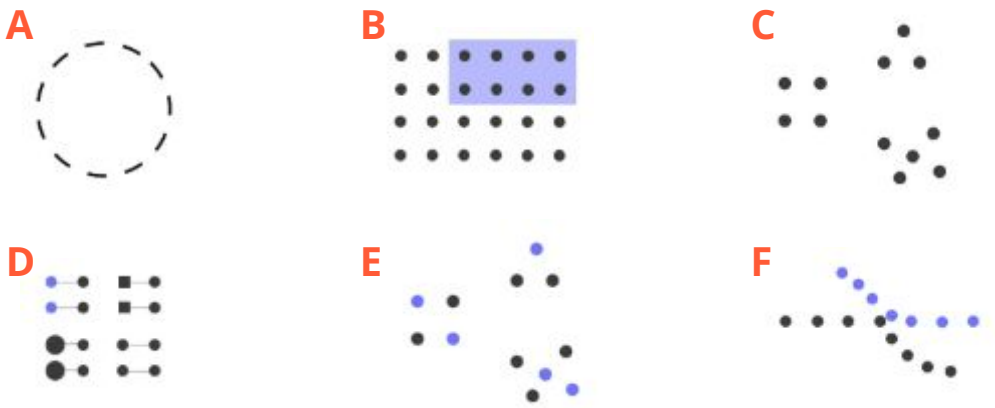


A picture is worth a thousand words...

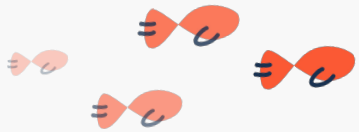
Telling our stories through graphs and charts!
Exercise Three: Gestalt Principles!

When it comes to understanding images, there are six principles that our brains follow. Can you match the image with the name?

1. Proximity
2. Similarity
3. Enclosure
4. Continuity
5. Connection
6. Closure



Introduction to Visualisation



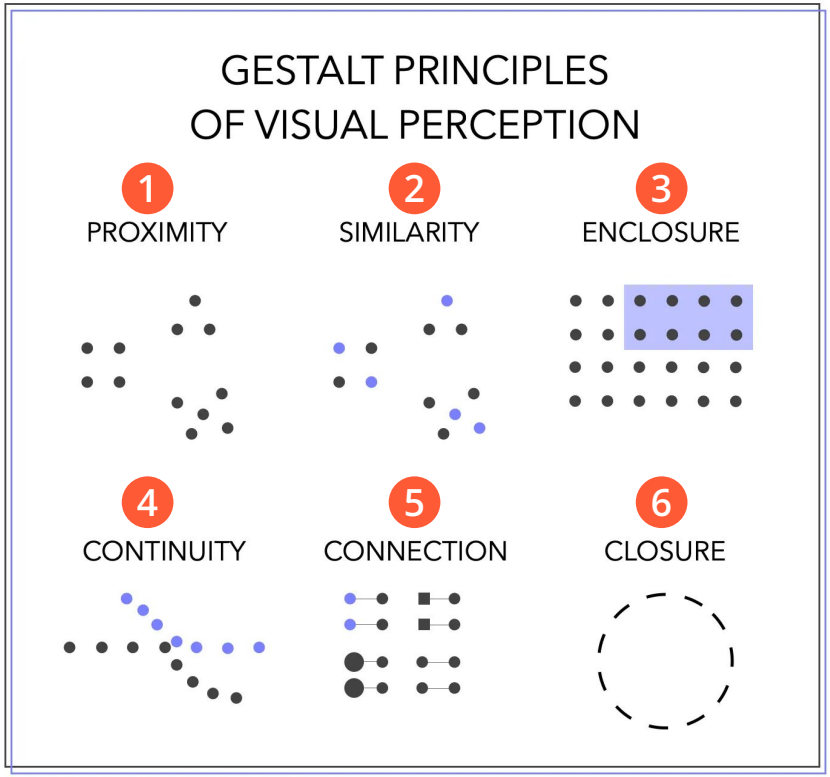
What are Gestalt Principles?

We group things that ...

1. are near each other,
2. are similar.
3. are enclosed,
4. continue each other,
5. are connected,

and

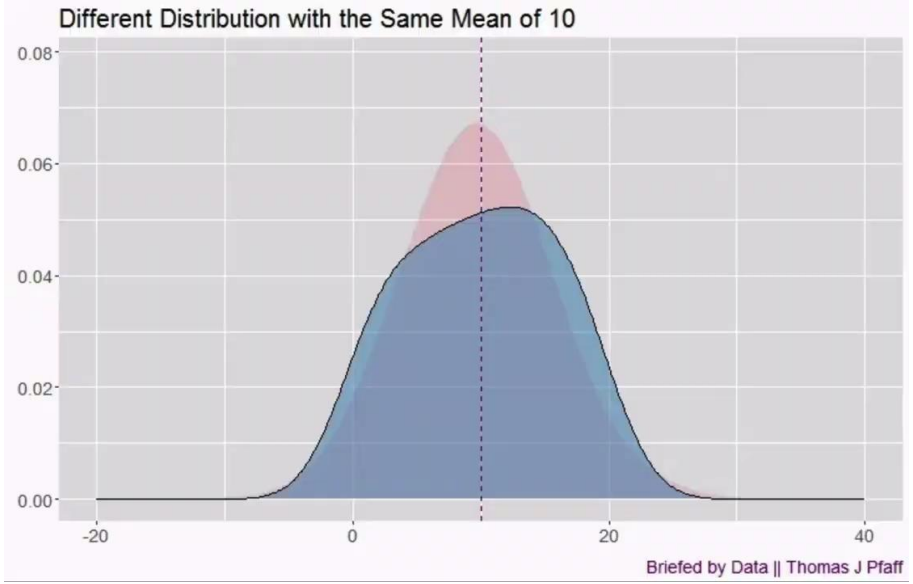
6. We close the gaps in what we see.



Why Do We Need Visualisation at all?

In short, because averages can be misleading.

See for yourself!



Source

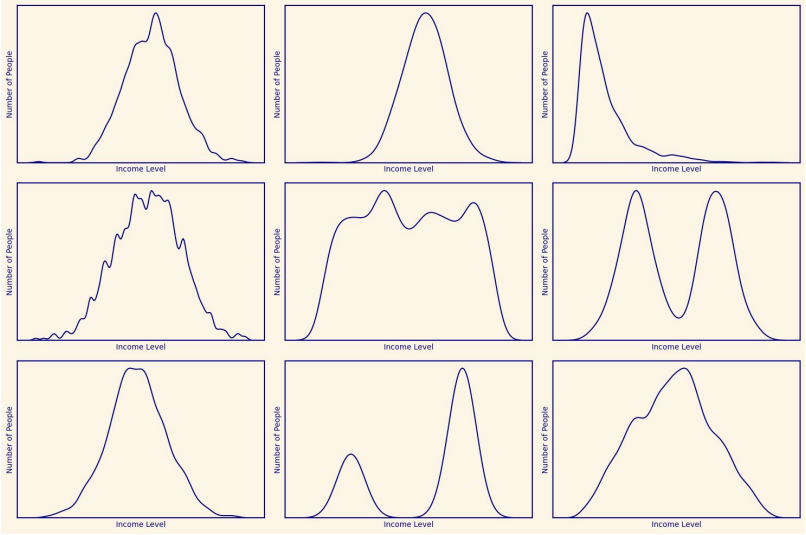
Why Do We Need Visualisation at all?

Let’s look at income level in a few societies.

Almost all of these graphs have the same average!

But are they equal? Of course not!

Which society would you like to live in?



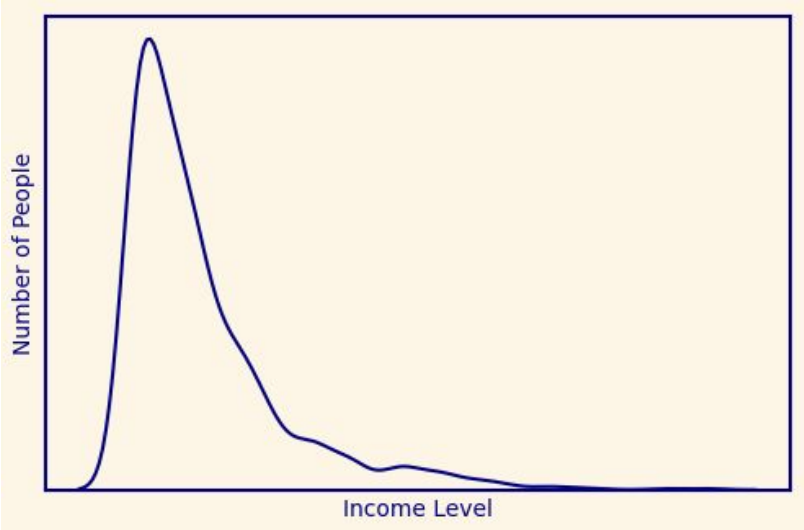
What Do We Want? Insight!

What does the graph tell us? One thing is:

"There are far more low income in this country than high income people."

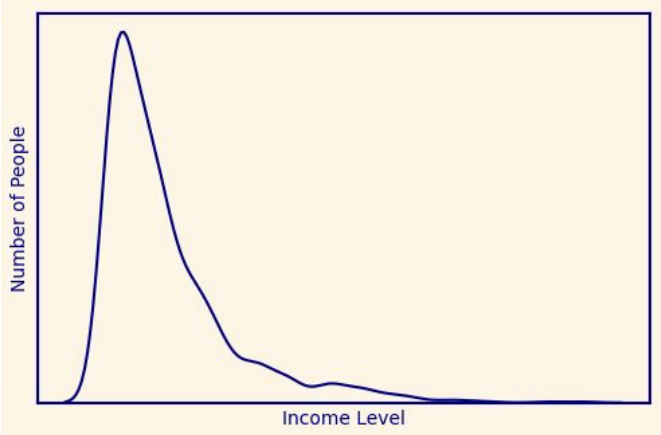
So now, we can

- ask more questions
- get more insights and
- reach a conclusion.

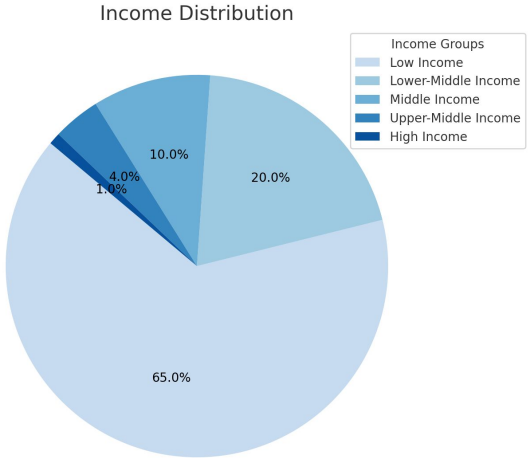


We Want To Share Our Insight!

Which visualisation conveys our message more clearly?



Or



It All Comes Down To This: Are We Exploring, Or Explaining?

Here is how it goes down:

1. Explore
2. Look at *all* possibilities
3. Find **insights**

→ **Exploratory**
For you
Goal:
understanding

And then

4. Tell your story to others
5. Guide them to the same insight
6. Lead to **actions**!

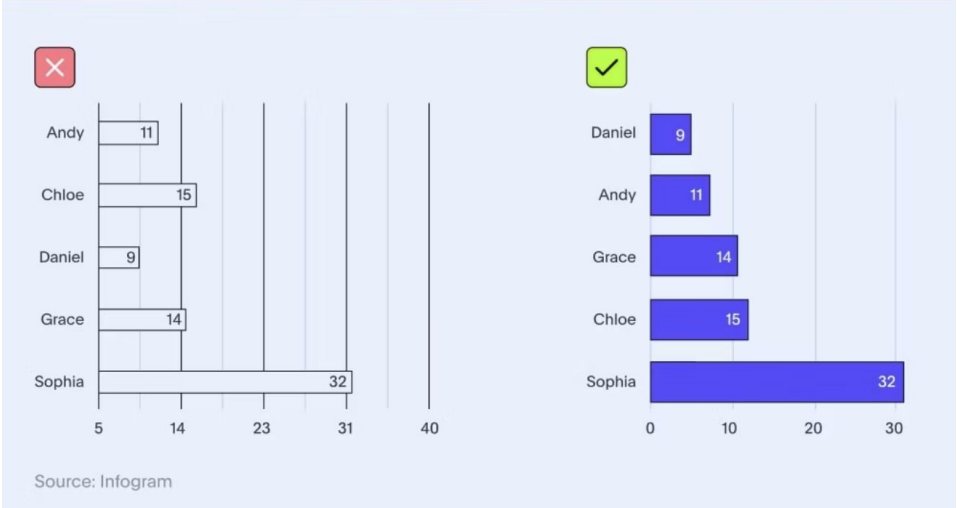
→ **Explanatory**
For others
Goal:
communicating

Good visual design isn’t decoration - it’s clarity.

Why Do Gestalt principles Matter for Data Viz?

Principles of good design:

- Highlight relationships
- Reduce cognitive load
- Make patterns visible instantly



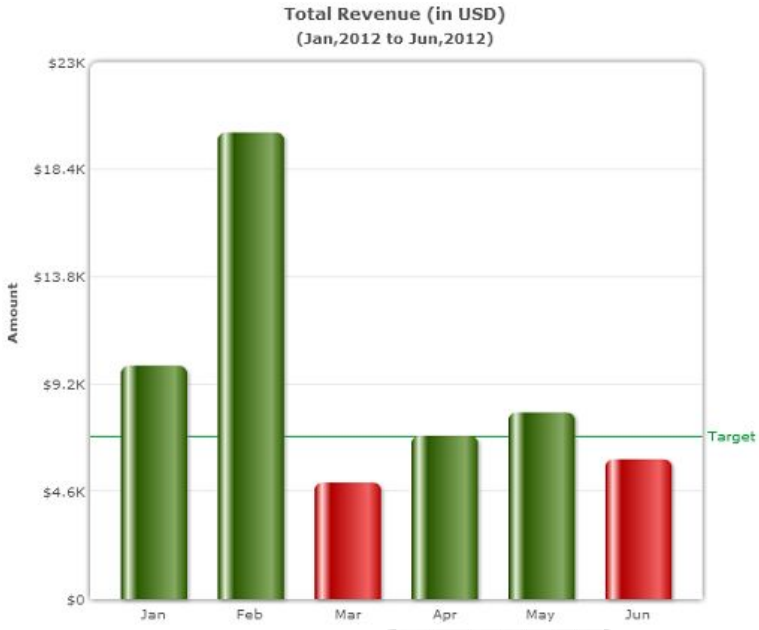
Do you get it?

What Makes a Viz Successful?

Design is successful when people *get it* - fast and correctly.

A successful visualisation is:

- ✅ Accurate
- ✅ Efficient
- ✅ Accessible *
- ✅ Purpose-driven



* Psst! What's wrong with with this visualisation?

So to wrap up...

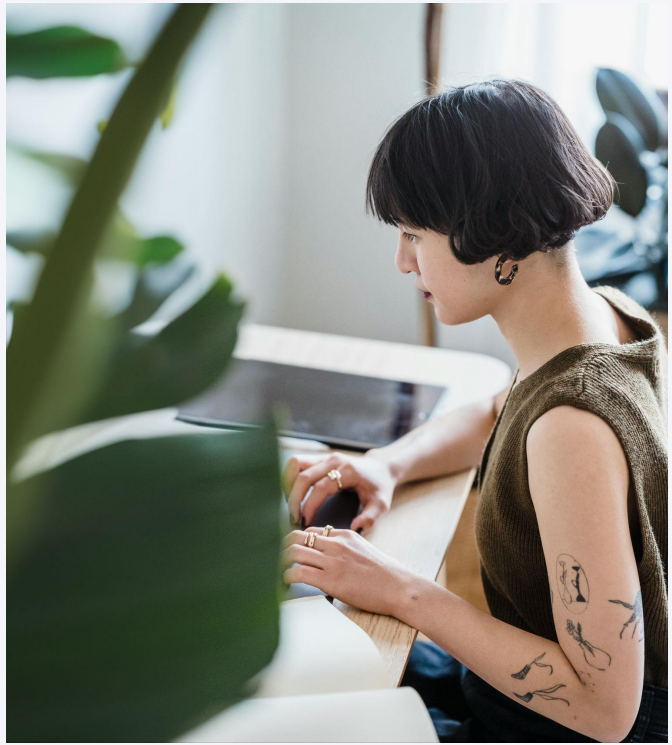
Applying the Checklist

To evaluate a visualisation, ask yourself:

- Is the message clear?
- Is the data represented truthfully?
- Is it readable and inclusive?
- Does it serve its purpose?

Do you have 32 seconds?

I Need You!



Your time is valuable!

To make the content the most effective in the least amount of time, I need to know:

- If it was useful,
  and
- How difficult it was

Please let me know here.
It will only take 32 second
(yes, I have timed it)

Now on to you...

Your Challenge!

It’s the iteration that turns sketches into stories!

Now that you know what to look for:

- Revisit the visual you created earlier
- Apply relevant Gestalt principles
- Clarify the message and purpose

In the live session, you will be checking that with your coach!

Good Luck and See You Soon!

